

PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	Data Flow Diagram	DATE	July 2026

Data Flow Diagram

The following describes the end-to-end data flow of the Credit Card Approval Prediction system, from raw data collection to the final prediction served through the web application.

Level 0 - Overview

- User -> Web Application (Flask) -> Trained Model -> Prediction Result -> User

Level 1 - Detailed Flow

- 1. Raw Data Sources: application_record.csv and credit_record.csv (Kaggle)
- 2. Data Preprocessing: cleaning, merging, feature engineering, encoding (Google Colab / Jupyter)
- 3. Model Training: Logistic Regression, Decision Tree, Random Forest, XGBoost trained on processed data
- 4. Model Selection: best model (Random Forest) serialized to model.pkl
- 5. Web Application: user submits form data via index.html
- 6. Flask Backend (app.py): encodes form input to match training format, loads model.pkl, calls predict()
- 7. Result Rendering: prediction (0 = Approved, 1 = Rejected) rendered on result.html

Data Flow Summary

Raw CSV Data -> Preprocessing & Feature Engineering -> Model Training & Evaluation -> Saved Model (model.pkl) -> Flask Application -> User Interface -> Prediction Output